



Probability Distributions

Key terms

KSBs: K13, S11, S52, S53, B1

discrete random variable, continuous random variable, probability distribution, probability density function, binomial distribution, normal distribution, continuity correction

3.1 Probability Distributions

Suppose we roll a fair 6-sided die
ones, twos, etc.

and record the number of

# Rolled	Frequency
1	12
2	9
3	11
4	10
5	7
6	11

If we went on rolling the die, we might expect
to come up on the die. In saying this, you would be using a perfectly reasonable

We could express this model as:

x	1	2	3	4	5	6
$P(X = x)$						

If we want to consider the case

we can write

which means “the probability that the die shows a 1.” This table shows the

The sum of the probabilities is

3.2 Discrete Random Variables

Definition 3.1 - Random Variable

A **random variable** (r.v.), usually denoted is a variable

where the sum of the

probabilities of all its possible values is

We write:

The range of values a variable can take is called the

Example 3.1. In the example given above,

Definition 3.2 - Discrete Random Variables

If a random variable
is a

we say it

Example 3.2. Consider the situation where we roll two fair six-sided dice. Let X be the random variable

The sample space here is the whole numbers between
given by:

The probability distribution of X is

x	2	3	4	5	6	7	8	9	10	11	12
$P(X = x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$					

For example, rolling a sum of 6 can be achieved

so it has a probability of and we say
as $1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1 = 36$.

We have

Exercise 3.1. (a) Let X be the denomination of coins put into a car parking meter. It is known that X follows the following probability distribution:

x	1p	2p	5p	10p	20p	50p	£1	£2
$P(X = x)$	0.02	0.02	0.05	0.07	0.08	0.16	0.4	0.2

(i) What is $P(X \geq 10p)$?

(ii) What is $P(5p < X \leq 50p)$?

(iii) What is $P(X \leq 25p)$?

(b) A random variable has the following distribution.

x	1	2	3	4
$P(X = x)$	42%	18%	14%	?

What is $P(X = 4)$?

3.2.1 Mean of a Discrete Random Variable

Definition 3.3 - Mean of a Discrete Random Variable

For a discrete random variable X , the **mean** of X ,
calculated as

can be cal-

For each of the values x that X can take, we

The mean is also referred to as the

of the random variable.

Example 3.3. For the 'sum of two dice' example above, the expected value is:

Note

If the values a discrete variable can take are

we still need $\sum P(X = x) = 1$ but we

We will mainly be working with variables that take numerical values.

Exercise 3.2. (a) The discrete random variable X has the following probability distribution:

x	-100	-50	-20	-10	0	100	200
$P(X = x)$	0.19	0.05	0.24	0.01	0.1	0.001	0.409

(i) Find the probability that the random variable takes a negative value.

(ii) Find the expectation of X .

(b) For the two probability distributions given in Exercise 3.1, find the mean of each.

3.2.2 Variance of a Discrete Random Variable

We can also calculate the variance of a discrete random variable.

Definition 3.4 - Variance and Standard Deviation of a Discrete Random Variable

For a discrete random variable X , the **variance** of X is given by

The **standard deviation** $\sigma(X)$ of a discrete random variable is

Note that here, instead of dividing by n or $n - 1$, we use $\sum P(X = x)$ (which is already a proportion out of n). Since the probability distribution gives us information about the whole population, this calculation corresponds to the one used in finding

Exercise 3.3. For the probability distributions given in Exercise 3.1 and 3.2, find the variance.

3.3 Continuous random variables

As well as defining probability distributions for discrete random variables, we can also consider

- which can take - and describe how the probability changes as
the value changes. In this case we call it a rather
than a probability distribution.

Definition 3.5 - Continuous Random Variable

A **continuous random variable** is one which

Definition 3.6 - Probability Density Function

For a continuous random variable, a **probability density function** is a

For continuous random variables, the probability of the variable taking any one specific value is The
value of the probability density function at a given point in the sample space can be interpreted as a
that the value of the random variable would be equal to that value - for example,

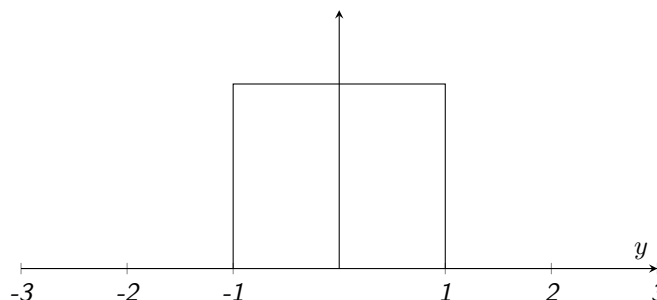
(It's not a graph showing what the probability is, but
which can be thought of as

While we can't use the function to find the probability of a particular value, we can instead use it to find

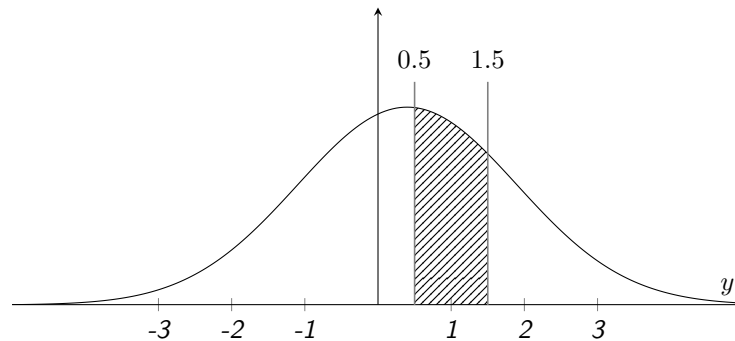
This can be calculated by finding the
We will see how this works later.

The area under the whole curve since this is the total probability of the distri-
bution.

Example 3.4. The plot below shows a density function with for
which all values occur with equal probability. Since the width of the
rectangle is we know the height must be since the area under the curve

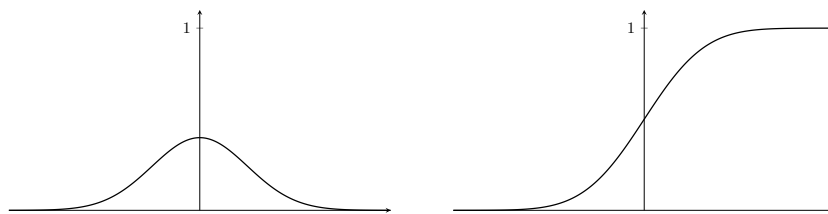


This plot shows a where values around μ are the
 most likely and the probability are the
 The region of values between $\mu - \sigma$ and $\mu + \sigma$ are shaded, and σ gives



It is also possible to plot the for
 which at each point the gives the

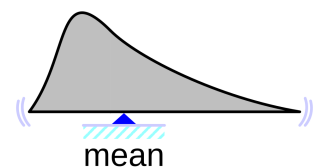
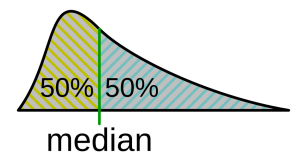
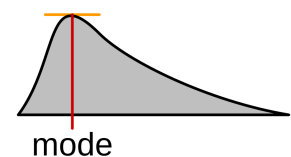
The cumulative probability curve and



We can also extend the definitions of statistics from discrete to continuous random variables, and they have reasonably intuitive geometric interpretations:

- The mode of a continuous random variable is the value of X that has the highest probability density.
- The median of a continuous random variable is the value of X such that the probability of X being less than or equal to that value is 0.5.
- The mean of a continuous random variable is the value of X such that the probability of X being less than or equal to that value is 0.5.

- if you were to make a physical model, it would balance on that point.



In practice, the only probability density functions you'll ever really need to work with are ones whose shapes match specific known mathematical functions, which we will see some examples of below.

If you have a set of data to work from, you don't need a probability density function, as you can instead - and in general, we won't be able to create a probability density function of any shape we want. What we can do is

or

and use that to make predictions.

3.4 Binomial Distribution

Definition 3.7 - Binomial Distribution

The **binomial distribution** is a $\text{Bin}(n, p)$ which measures how likely it is that, if we perform n trials, which succeeds with probability p we have X successes. If a random variable X has a binomial distribution, we can write $X \sim \text{Bin}(n, p)$ or

Note

We use the symbol $\sim$ to mean

There are four criteria we need for a situation to be modelled using a binomial distribution:

- There are only n trials (i.e. the outcome of one trial doesn't affect any other trial)
- Each trial is independent
- There is a fixed probability p of success
- The probability of success is the same for every trial

Example 3.5. Suppose I flipped a fair coin 5 times and measured the number of times it lands on heads.

- There are 5 trials
- Each trial is independent
- There are a fixed number of trials
- The probability of success is the same for every trial. There is always a 0.5 chance of heads for each flip.

The four conditions are satisfied,

So if X is the number of times the coin lands on heads, then

Exercise 3.4.

- (a) For each of the situations below, state which of the conditions of being modelled using a binomial distribution it fails to satisfy.
- (i) A coin is flipped until it lands “tails”.
 - (ii) An archery student fires 13 arrows, and the number of times they hit the target is recorded.
 - (iii) A group of 15 friends arrive at the college, and the number of students who get to class late is recorded.
 - (iv) A die is rolled and the score on the die is recorded.
- (b) Write down the expression for the binomial distribution for each of the examples below, by determining the values of n and p in each case.
- (i) In a class of 30 students, it is expected that 22% of them will have brown eyes.
 - (ii) In one week, each day from Monday to Friday, Noel catches a train from Piccadilly station. He claims that the probability that the train arrives on time at the station is 0.4, and that the train's arrival time is independent from day to day.

3.4.1 Finding probabilities using a binomial distribution

Example 3.6. A fair coin is flipped 5 times, and let X be ‘the number of heads’. So What is $P(X = 3)$?

There are 10 ways to get exactly three heads: HHHTT, HHTHT, HHTTH, HTHHT, HTHTH, HTTHH, THHHT, THHTH, THTHH and TTHHH. Since then the

probability of getting (in a given order) is Since there are we have

Example 3.7. A six-sided fair die is rolled 5 times, and let X be ‘the number of times I roll a 6’. The probability of getting a 6 is and for each roll I either roll a 6 or not a 6, so we know that What is the probability that I roll a 6 exactly once?

Rolling a six exactly once can occur it can be the

The probability of rolling a 6 is and the probability of

So the probability of rolling (in a given order) is $\left(\frac{1}{6}\right) \times \left(\frac{5}{6}\right)^4$. Since this can happen in we have

In general, the calculations may be too complicated to do this by hand, so we can use computational tools. Later, we will see other ways to work with binomial distributions that allow us to compute it more easily.

Function: BINOM.DIST() (E/S)

The `=BINOM.DIST()` function will calculate a binomial distribution. It takes four parameters:

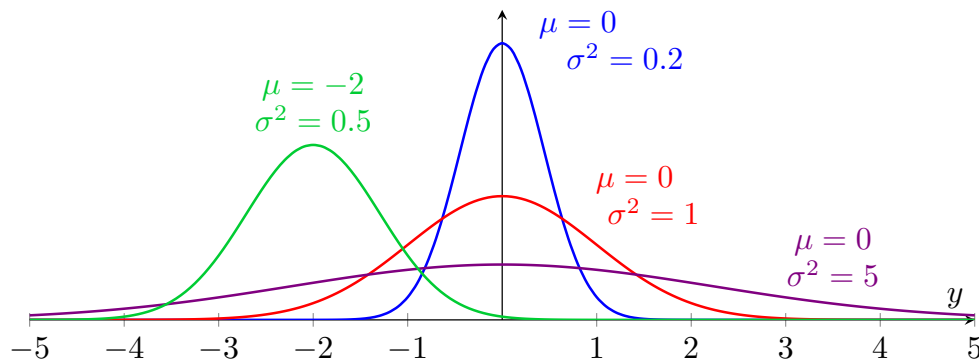
- k , which is the number of successes you wish to find the probability of
- n , the number of independent trials
- p , the probability of success on each trial
- a fourth parameter to specify whether you want the cumulative distribution (TRUE) or not (FALSE); for the probability of 'at most k successes' you want TRUE, and if you set it to FALSE it will tell you the probability of 'exactly k successes'.

3.5 Normal Distribution

Definition 3.8 - Normal Distribution

The **normal distribution**, sometimes also called a $\text{Gaussian distribution}$ is a probability distribution for a $\text{continuous random variable}$ which is commonly used as a model for many real-world distributions. If X is a random variable that follows this distribution, and has a mean of μ and a standard deviation of σ we write:

The $\text{standard normal distribution}$ is the version of this distribution with a mean of 0 and a standard deviation of 1 denoted

**Note**

Note that the two numbers in the bracket $N(\mu, \sigma^2)$ are the mean and variance even though we talk about normal distributions in terms of the standard deviation. It's conventional to leave the expression as $N(\mu, \sigma^2)$ rather than evaluate it as a number - so for a distribution with a mean of 2 and a standard deviation of 5, we'd write

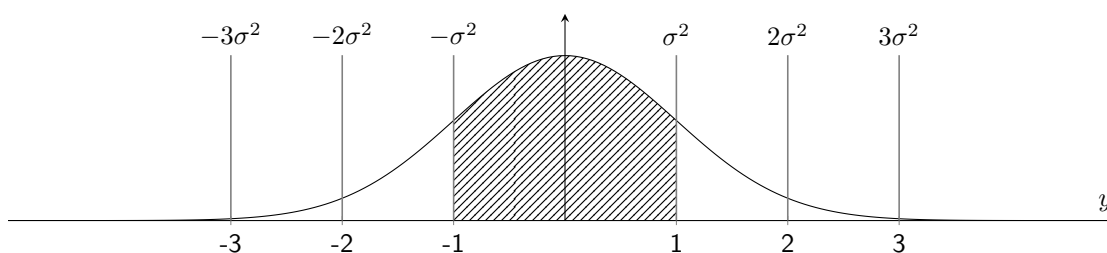
The probability density function for the normal distribution takes the shape of a symmetrical curve, sometimes also called a bell curve which has the equation:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

You don't need to know this equation - but it's worth noting that the curve is entirely specified by the mean and standard deviation of the distribution.

Varying the mean will move the mean along the horizontal axis, and changing the variance will change the shape of the curve so it's taller/narrower or shorter/wider - to maintain an area of 1

- The curve is always symmetrical with the highest point at the mean
- The mean and median of the distribution are the same
- About 68% of the area under the curve (shaded below) lies within 1 standard deviation of the mean.
- About 95% lies within 2 standard deviations of the mean.
- Almost all the area 99.7% lies within 3 standard deviations of the mean.



Since the shape of the curve depends on the mean and standard deviation we can use it as a way to describe points on the curve, in terms of standard deviations from the mean

Definition 3.9 - z-score for a normal distribution

For a given value x the random variable could take, the **z-score** is a way to

normal distribution with mean μ and standard deviation σ :

Given a

Subtracting μ from the value of x tells us

dividing by σ

so it fits the curve we're looking at. This allows us to find the point on the curve corresponding to the value of x .

Theorem 3.1 - Sum of two normal distributions

Let X and Y be independent random variables that are normally distributed.

Then their sum is

That is, if

and

then

Example 3.8. A baker is weighing ingredients for a cake by eye. Let

represent

the amount of flour weighed out in grams, and

represent the amount of sugar.

Assume the two are independent.

The total weight of flour and sugar can be expected to follow

3.5.1 Finding probabilities using a normal distribution

We have described the probability as being related to the and there are mathematical techniques for finding the area under a curve between two points; however, when the curve is not a plot of a simple-to-describe curve like $f(x) = x^2$, these techniques are not easy to apply. Instead, the area is pre-calculated for a large number of values, and this information can be looked up using a table or function.

The traditional method (and the one we'll use in the exam) is to consult a one given below.

like the

To use this table (and others like it), you need to calculate the

you'd

like to know the probability of, for the

Recall that this can be calculated as

If this value is positive, you can

the first digit after the decimal

place tells you

then the second digit after the decimal place

For example, to look up a z-score of

we'd go to the row corresponding to

and the column for

as adding these two numbers together gives

The number you get from the table will tell you the

(which is the area under the curve

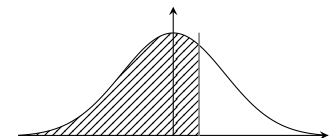
Note that when $z = 0$

the probability is since exactly half the curve lies to each side. If we move

away from the mean, the value

(as it includes the left half of

the curve, as well as the part between the point you're at and the middle).



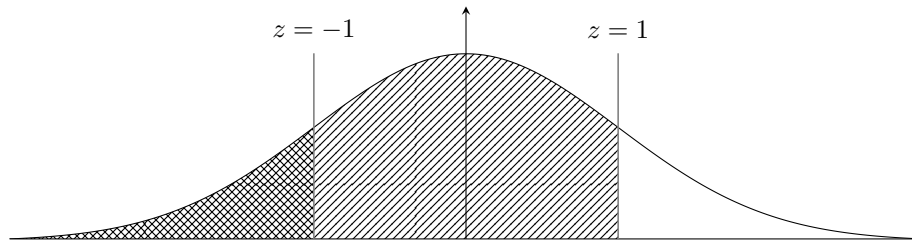
If the z -score is negative, we use the fact that the distribution is

to find the area to the left.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999							

A copy of this table will be included in the exam paper.

Example 3.9. We have calculated the z -score to be $z = -1$. Since this is $z = 1$ we can instead look up the z -score $z = 1$ in the table, which will give us the same value. This is



Function: NORM.DIST() (E/S)

The `=NORM.DIST()` function will calculate a normal distribution. It takes four parameters:

- x , which is the value you want to calculate the probability for
- μ , the mean of the normal distribution you're studying
- σ , the standard deviation of the normal distribution you're studying
- a fourth parameter to specify whether you want the cumulative distribution (TRUE) or not (FALSE); for the area under the curve, you want this to be TRUE, and if you set it to FALSE it will just tell you the height of the curve at that point.

Exercise 3.5.

1. Let X be the distribution of the heights of men from Albania. We assume that X is a normal distribution, with a mean of 174cm and standard deviation of 11 cm. What is the probability that if we spoke to a random Albanian man, he will be taller than 160cm?
2. Let $X \sim N(60.1, 9.392)$.
 - (i) Find $P(X < 58.9)$.
 - (ii) Confirm that $P(X > 57.8)$ is 0.5967.
 - (iii) Find $P(X \geq 70)$.
 - (iv) Confirm that $P(54 \leq X < 58.4)$ is 0.1702.

3.5.2 Applications of the normal distribution

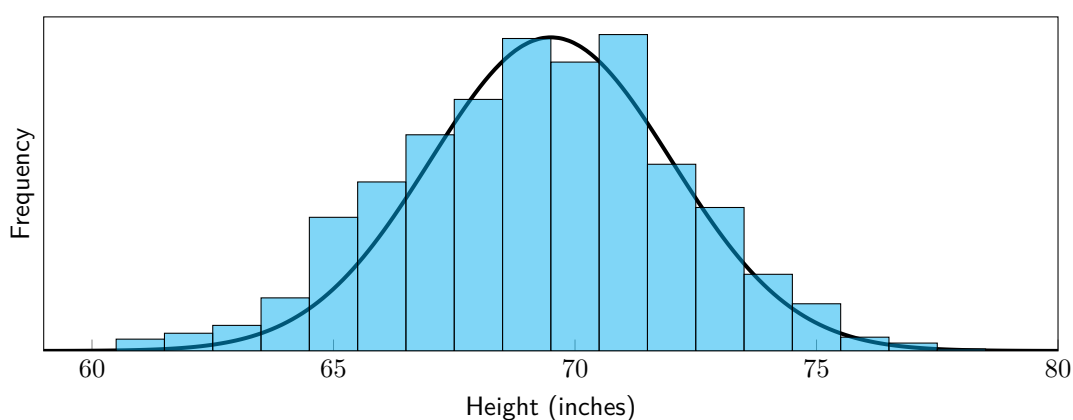
We have been discussing the normal distribution as a
but this can be thought of as equivalent to

and all the same maths applies.

Many real-world datasets are extremely well approximated by the normal distribution - in particular, things which
are affected by

For example,

fit a normal curve (below), as do



All of these types of data follow a similar shape: most values are
and there are

This means the normal distribution can be used to model many real-world datasets. Once it has been established that the distribution is normal, you can use the
to construct a curve which represents a good approximation to the distribution you would expect for the whole population.

Since the normal distribution is an idealised mathematical curve, it continues on in both directions,

This means that it would predict a tiny but non-zero probability of

This would be such a small probability that in practice you don't need to worry about it, but it's worth remembering that this is a model and not actual data.

Some real-world data does not follow a normal distribution - anything which involves

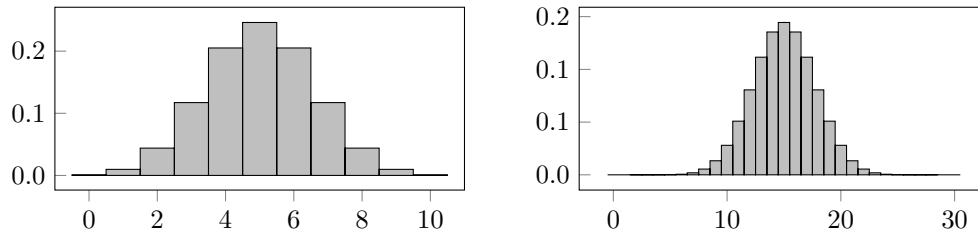
(where many observations might be zero, with a curve that drops off). It's important to check, by looking at the shape of the data, which distribution is the right one for your model.

3.6 Approximating a Binomial Distribution using a Normal Distribution

The plot below shows

when a

fair coin is flipped 10 times (left), and 30 times (right).



The shape of the binomial distribution

As the number of trials increases, the approximation gets better, and it works best when

Definition 3.10 - Normal Approximation to a Binomial Distribution

For $n > 20$, we can **approximate a binomial distribution using a normal distribution**, with

If $X_B \sim B(n, p)$, then

is an approximation for X_B .

Note

The requirement that $n > 20$ is a useful rule of thumb, but larger n give a better approximation.

When we approximate a discrete distribution by a continuous one, under certain conditions we must apply a continuity correction to make the approximation better.

Definition 3.11 - Continuity Correction for the Normal Approximation for Binomials

When approximating $X_B \sim B(n, p)$ using $X_N \sim N(np, np(1-p))$, if the values of np and $n(1-p)$ are
we must apply a **continuity correction**:

This correction means that if we are interested in e.g. the probability of the discrete value being we include the whole of the bar centred at 6, which has width 1.

Example 3.10.

- Consider $X \sim B(5, 0.2)$.
- Consider $X \sim B(25, 0.1)$. We can approximate this by

Since

In order to find $P(X < 3)$, we can calculate

Looking this up in the normal distribution table gives

- *Consider $X \sim B(100, 0.25)$.*

To find the probability of $19 \leq X \leq 23$, we must

and find

The normal distribution table tells us that

Exercise 3.6.

- (a) *Let $X \sim B(60, 0.25)$. Using the normal approximation to the binomial, find $P(X \leq 13)$.*
- (b) *Let $X \sim B(56, 0.64)$. Using the normal approximation to the binomial, find $P(30 < X < 40)$.*

